



UTM

UNIVERSITI TEKNOLOGI MALAYSIA

PROJECT 1: DISABILITY CARE SOFTWARE

GROUP 3

SECV2113-01 HUMAN COMPUTER INTERACTION

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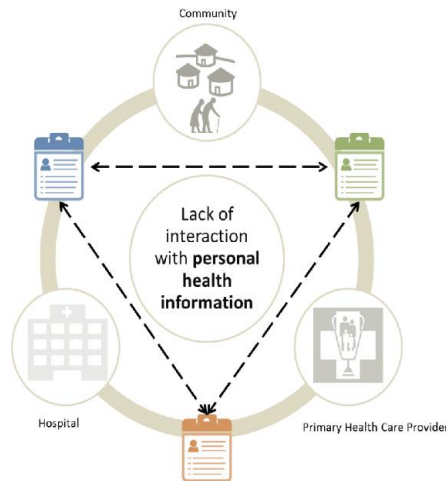
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1.0 Introduction

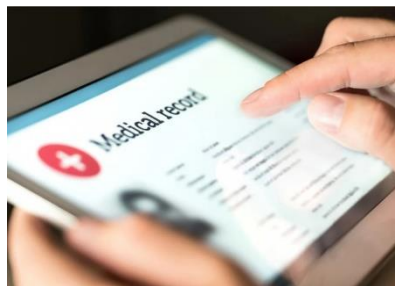
In today's healthcare landscape, individuals with disabilities encounter significant challenges in managing their health and life due to lacking proper healthcare support and facing inadequate systems. An innovative solution is required to address the special needs of these vulnerable people. To address the many problems that people with disabilities encounter, such as incomplete mental health treatment, inconsistent drug administration, and fragmented healthcare information, our group has proposed to build Disability Care Software. In connection with SDG 3, which is good health and well-being, our integrated approach seeks to provide complete solutions, such as integrated healthcare record management, to assist individuals in maintaining a healthy lifestyle. Our solution aims to enable people with disabilities especially those who suffer from mental and physical health issues and elderly people to lead healthier and more fulfilling lives.

2.0 Problem Clarification

Problem 1: Lacking of personal health information



Lacking of personal health information becomes the main problem for individuals with disabilities, especially elderly people and those with mental and physical health issues. These individuals often experience fragmented healthcare information, hindering their ability to access and manage their personal health information efficiently. Moreover, this fragmented healthcare system arises from several sources of health data, such as electronic health records, paper-based documents, and personal health tracking devices.



The primary usability problem is the absence of a centralized platform for handling personal health data. Individuals with disabilities must navigate numerous sources and formats of health data in the absence of a unified system, leading to confusion, inefficiency, and record-keeping problems.



When it comes to user experience, users might find it difficult to update the personal medical problem manually, track and monitor their health condition completely, which could cause delays in treatment, miss opportunities for early intervention, and result in worse than ideal treatment outcomes.

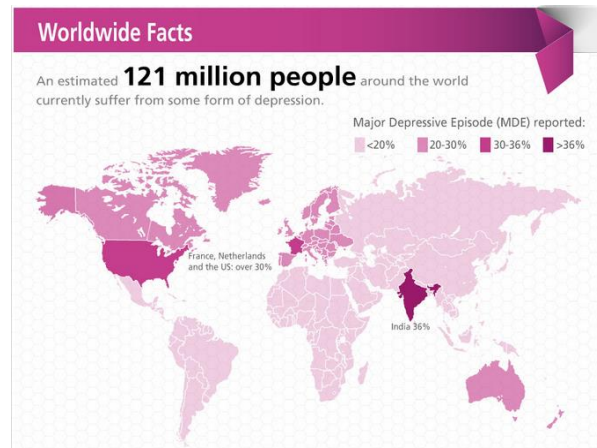
Problem 2: Lacking of awareness on mental health maintenance

In many societies, there exists a lack of awareness regarding the significance of mental health maintenance. This ignorance often stems from deeply ingrained cultural beliefs and norms, particularly evident in Asian households where discussing mental health is often taboo. Adolescents dealing with mental health issues frequently get ignored when attempting to seek professional help from their parents. Instead of receiving the support they need, their concerns may be brushed aside, undermining their well-being and worsen their conditions.

This societal neglect of mental health care can have devastating consequences, particularly for vulnerable individuals already grappling with mental illnesses. When individuals are denied the opportunity to address their mental health concerns openly and receive appropriate treatment, their conditions may escalate, leading to harmful behaviours such as self-harm. The stigma surrounding mental health further compounds the problem. The common conception is that there's something wrong with a person as a human being if he or she suffers from a mental illness.

Moreover, mental health is perceived as a luxury good. If you suffer from depression, it means you are just a whiny person with all their basic needs satisfied. This mindset is very wrong but it is still very common to see many people having this wrong mindset nowadays. If

mental health is the most neglected health problem in the developing world, depression is the single most prevalent mental illness. Depression affects women at twice the rate of men.



Problem 3: Difficulty in Medication Management



For many individuals dealing with mental health issues, physical ailments, or advancing age, managing medications presents a significant challenge. Limited financial resources often stand in the way of accessing essential healthcare resources, making it difficult to afford medications or seek necessary treatments. Additionally, for those with disabilities or mobility impairments, getting to healthcare facilities can be inconvenient or even impossible, further exacerbating the problem. This lack of access not only impacts their ability to obtain vital medications but also prevents them from accessing important government benefits designed to support their health needs.

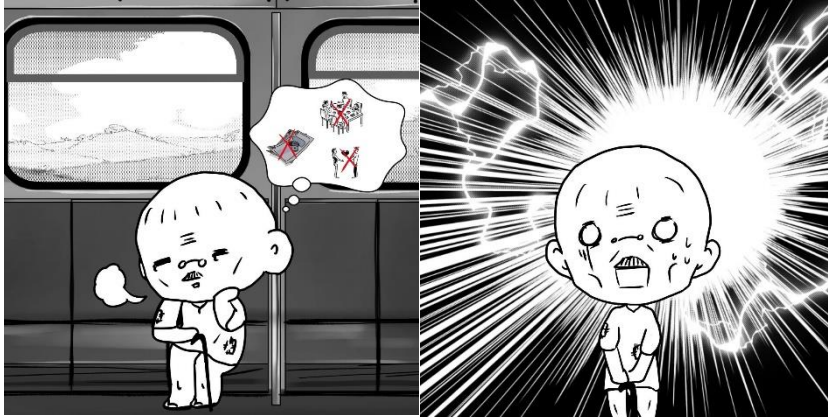


Consequently, individuals facing these obstacles often find themselves struggling to maintain a consistent medication regimen, leading to worsening health conditions and decreased quality of life. Without proper access to healthcare resources and support, they may experience heightened health risks and increased vulnerability to complications. Addressing these barriers to medication management is essential to ensure that vulnerable populations receive the care and support they need to manage their health effectively and improve their overall well-being.

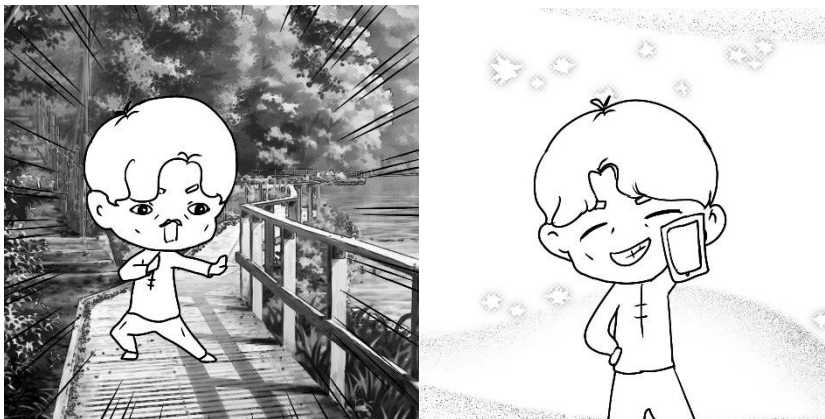


3.0 Illustration to Problems

Problem 1:



Mr. Harris, a lonely elderly man, he had no family to call his own, no true love to share his days with, and as the years had advanced, his savings dwindled, leaving him with little more than the bare necessities.



As he wandered along the winding pathways, lost in his thoughts, he noticed a man was doing exercise, basking in the warm sunlight. Despite being of a similar age to Mr. Harris, the man seemed remarkably healthy and vibrant.

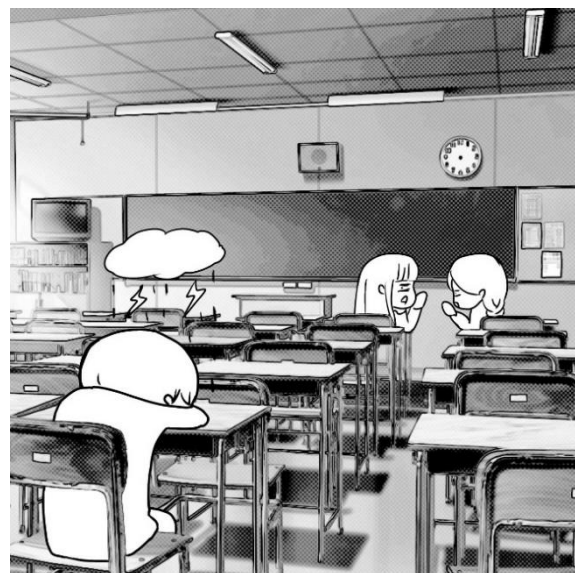
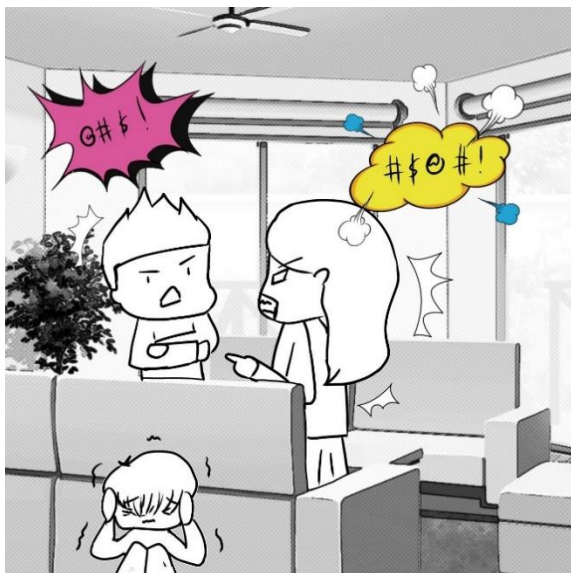


To his surprise, the man shared with him the secret to his well-being: a mobile app that provided free access to doctor consultations and health services. With a grateful smile, the man explained how the app had transformed his life, allowing him to address his health concerns without the burden of exorbitant medical bills.

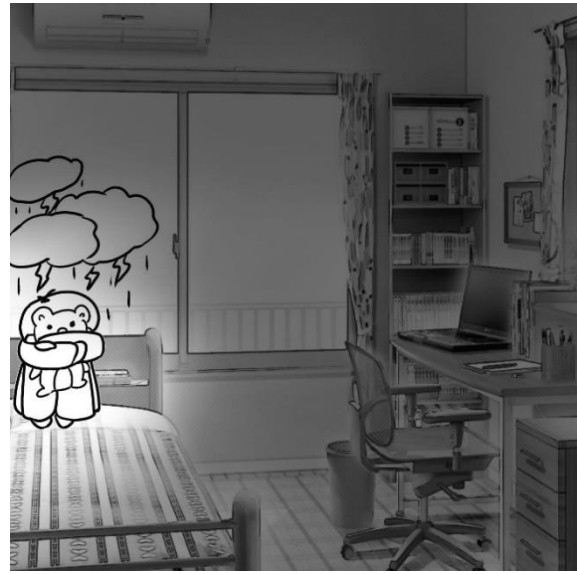


As Mr. Harris continued his walk through the park, he encountered another unexpected sight, an elderly woman laden with shopping bags, her face radiant with satisfaction. With a twinkle in her eye, the woman revealed her secret: a mobile app that alerted her to government assistance programs and free services available to seniors. Through the app, she had been able to claim various benefits and subsidies, easing the financial strain of her golden years.

Problem 2:



Ethan parents' constant arguing and simmering anger filled the air of their home, leaving Ethan feeling like he was walking on eggshells every day. The other children at school sensed Ethan's unease, and they kept their distance, unsure of how to approach him. With no friends to confide in, Ethan bottled up his emotions, afraid to burden anyone with his troubles.



To make matters worse, Ethan's mother had high expectations for him. She pushed him relentlessly to excel in school, to be the best at everything he did. But under the weight of her expectations, Ethan felt suffocated, like he could never measure up. One evening, as the familiar sounds of his parents' arguing reached a crescendo, Ethan couldn't take it anymore. He fled to his room, locking the door behind him, seeking refuge in the darkness. Tears streamed down his cheeks as he collapsed onto his bed, the weight of his despair crushing him.

Problem 3:



One morning, Mary was feeling sick, but she doesn't have transportation and her family are busy, so she decided to endure the pain herself.



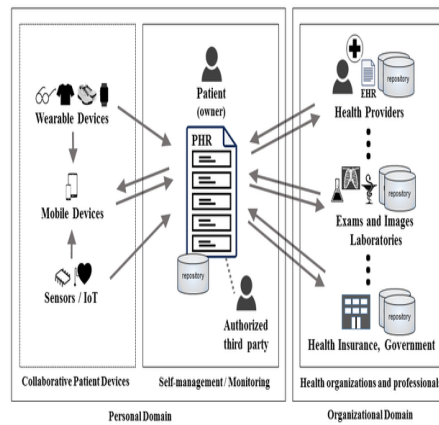
By evening, Mary found herself struggling to even move her hand. But as she wanted to reach for help, the world began to spin around her, and everything went black. When Mary's sister arrived home later that evening, she found her sister lying unconscious on the wheelchair in the kitchen.



At the hospital, the doctors ran a medical body check up on Mary, trying to determine what had caused her sudden collapse. After hours of waiting, the results came back, Mary had a serious underlying medical condition that required monthly check-ups to monitor. The news was a shock to Mary and her family, Mary also realised that she couldn't keep ignoring her health issues again.

4.0 Proposed Solutions

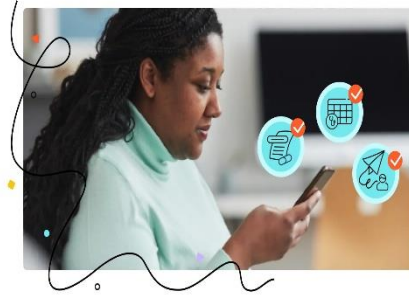
Solution 1: Design an integrated healthcare record management system



The first problem is fragmented healthcare information. We plan to design an integrated healthcare record management system that provides a centralized platform where users can securely store, access, and manage their personal health information from various sources, including electronic health records, paper-based documents, and personal health tracking devices.



The app should include important medical tests, such as blood pressure tests and cholesterol screenings, within the app serves as a crucial component of our integrated healthcare record management system. These tests provide vital indicators of an individual's overall health status and help to establish a comprehensive understanding of their well-being.



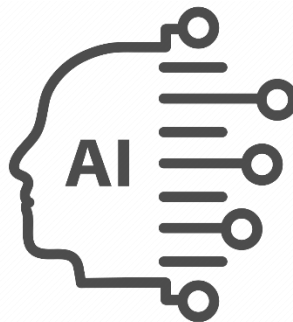
After users do medical checkups, they need to either snap a photo of the document result or send the lab report in PDF format to the system or update other personal medical background manually. The system will categorize all relevant information in one place, facilitating better decision-making and care coordination.



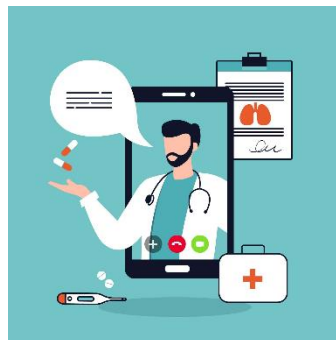
We will also ensure that users have a comprehensive view of their health information by designing an easy-to-use interface for the elderly to update their information. The system will analyse the users' medical results to provide them with advice and actions, ensuring that opportunities for early intervention are not missed.

Solution 2: Creating a Online Chatting Platform

The solution for problem 2 is creating a chatting platform with AI characters in our app. This not only facilitates communication between users but also serves as a medium for sharing daily experiences and seeking support. This feature enables individuals to connect with others, share their joys, struggles, and seek advice in a supportive virtual environment. By leveraging the platform's AI characters, users can engage in conversations that provide companionship and understanding, helping to alleviate feelings of isolation and loneliness caused by the mental illness.

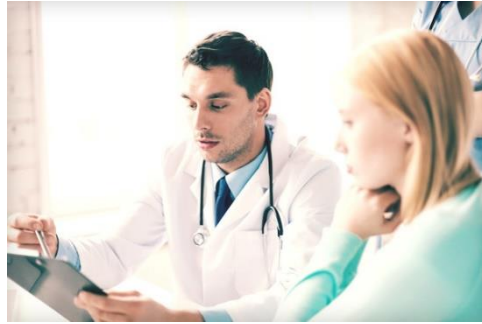


Moreover, there is also an online consultation feature which bridges the gap between individuals in need of mental health support and professional therapists or doctors in our app. This accessible feature empowers individuals, especially those facing resistance from family members in seeking help, to connect with trained professionals confidentially and conveniently. For adolescents unable to access traditional therapy due to parental barriers, online consultations allow them to address their mental health concerns independently.



In addition to enhancing accessibility, online consultations also offer a cost-effective alternative to traditional therapy sessions. By eliminating the need for physical travel and overhead costs associated with in-person visits, online consultations make mental health care more financially feasible for individuals and families. This affordability expands access to essential mental health services, making sure that financial constraints do not stop individuals from receiving the support they need to manage their mental well-being effectively.

Solution 3: Provide Free Online Consultation & Reminders



To address the healthcare needs of individuals dealing with mental health issues, physical ailments, and elderly individuals, we propose a comprehensive solution leveraging technology. Our app offers a range of features designed to make managing health easier and more accessible for everyone. For those with mental health concerns, the app provides free medical checkups and appointment reminders, ensuring they receive timely care and support. Additionally, AI notifications not only remind users to take their medication but also alert family members, fostering a supportive environment. Furthermore, the app automatically notifies users of available health resources, empowering individuals to take proactive steps towards their well-being.



Medical appointment reminders and management features allow users to schedule appointments, receive disease information, and access essential healthcare tips, promoting better self-care and disease management. Lastly, for elderly individuals, the app includes tailored features such as medication reminders and benefits assistance, making it easier for them to stay healthy and independent. By providing these comprehensive solutions, our app aims to empower individuals of all ages and health conditions to lead healthier, happier lives.



5.0 Target User

Based on the problems and solutions proposed above, there are three major target users for our services, which are:

- People with Mental Health Issues
- People with Physical Health Issues
- Elderly Individuals

i. People with Mental Health Issues

Our solution aims to assist individuals facing mental health issues in managing their healthcare needs effectively. This group of users will always facing challenges in managing their mental well-being due to factors like low self-esteem, lack of support, and difficult to get appropriate care. Therefore, our app can address their needs in several ways. Firstly, we offer an integrated healthcare record management system, allowing users to make sure their treatment plans are comprehensive and effective. This will include tracking their medical regimens, therapy process, and progress note. Additionally, our app provides a platform for communication, which can be a supportive community where individuals can share experiences. This social support aspect is essential for the people with mentally problems because they can get emotional support there. Furthermore, our app provide online consultant service. This feature important for those who are having metal issues because they can meet online counsellor and get mental support.

ii People with Physical Health Issues

For those who are having physical health issue like diabetes, cardiovascular diseases or stroke, they will always struggle with managing their conditions and maintaining medical schedules. For individuals with chronic condition like diabetes, hypertension, and obesity, our Integrated Health Record Management feature allow them to input their daily health data like blood pressure, glucose level, heart rate and any other relevant data. Then the system will analyze their daily health data and provide personalized insights and suggestions, so they can have more detailed idea on their daily meal to control their health data. Moreover, some of them may have mobility limitation, this app provide home doctor service which provide convenience and personalized care. What's more, they can also access to the online

consultant service to contact with experienced doctor to get more detail on their treatment plan.

iii. Elderly individual

The main problems that elderly faced will be managing chronic conditions, memory decline, and navigating the complex healthcare systems. Furthermore, some of them may feeling alone at home. Our software provide some service to address their need. First of all, the elderly can get benefit from the integrated healthcare record management system, which provide recommendation and suggestion on health by analyze their medical history and prescriptions. Moreover, our app provide the online community for social connection, aiming to help elderly overcome loneliness by allow them to chat with someone easily. Also, they can know others who are using the same app through this community. Lastly, most of the elderly will have memory decline, our app will push notification to the elderly and also the person linked so that they will not miss any medication schedules or any important checkup and consultations with healthcare professionals.

By addressing the unique needs of each target user groups, our app aims to enhance overall health outcomes, improve quality of life, and promote independence and well-being among individuals with disabilities.